

CSC-150 – Object Oriented Programming Syllabus

General Information

Course Number	CSC-150
Credit Hours	3+1 (Theory Credit Hour = 3, Lab Credit Hours = 1)
Instructor	

Course Objectives

This course is designed to provide students base for big programming to be able to develop the complex applications in manageable way that will be more organized, easy to understand, and easy to develop and for stated purpose object oriented paradigm will be used and for that java as a language is selected.

Java is widely used language nowadays and according to oracle more than 10 billion devices are using java in different application domains. This course is designed to target console and desktop (GUI based) applications majorly other than that all the constructs that will be discussed in this course will benefit students for development of java based web applications and the same java based libraries are used for development of mobile applications for well-known mobile operating system android.

Catalog Description

CSC-150

Assessment

MID TERM EXAM	40
Assignment / Quizzes	10
Presentation / lab Task	10
Final Exam	40

Text Book

1. Java the complete reference, 10 th edition	By Herbert Schildt
2. Thinking in java, 4 th edition	By Bruce Eckel
Reference Books	
1.	The java language specification By James gosling, Bill joy, Guy steele
2.	Ivor Horton, Beginning Java 2, 5th Edition
3.	Walter Savitch, Absolute Java, 5thEdition/6th Edition

COURSE CONTENT

Sr # Week	Topic	Topic Details
1	Moving C to Java	1. Procedural vs. OOP 2. Why OOP? 3. History of Programming Languages, C++, Java, OOP 4. Why JAVA? 5. Buzzwords/ Keywords 6. JDK, JRE and JVM
1	Introduction to OOP	1. Programming Paradigm 2. Abstraction 3. OOP Principles (Encapsulation Inheritance & Polymorphism)
2	Introduction to Java	1. First Java application/Short Examples 2. Two Control Statements (if and for) 3. Using Block Code 4. Lexical Issues 5. Programming Style 6. Introduction to Applets, Creating Applet and Embedding, 8. Java Documentation
2 & 3	Fundamental Elements of Programing Language	1. Fundamental Elements of Java (Datatypes, Variables & Array) 2. Primitive Types 3. Operators (Arthematic, Logical & Relational) Pre & Post Increment & Decrement 4. Escape Sequence 5. Scope and lifetime of Variables 6. Type Conversion and Casting 7. Selection Structures (If else, Switch) 8. Repetition (For Loop, Do While, While Loop) 9. Arrays (One Dimension & Two Dimension) 10. Methods(Static and Non Static)
4	OOP Concepts In Java	1. OO software development life cycle 2. Class 3. Class Definition 4. Class Declaration 5. component of class 6. Example of class
5	Classes and Objects	1. Class & Object 2. Method & Behavior 3. State & Message
5	Class Relationships	1. Association 2. Aggregation

7 & 8	Encapsulation: Closer look at Classes, Methods, Access Controllers & References	1. Methods 2. Overloading Methods 3. Overloading Methods – Automatic Conversion 4. Overloading Constructors 5. Using Objects as Parameters 6. Object to its Constructor as Parameter 7. A Closer Look at Argument Passing 8. Arguments in JAVA 9. Call by value 10. Call by Reference 11. Returning Objects 12. Access Modifiers (Specifiers: public, private protected) 13. Understanding static and final 14. Nested Classes 15. Inner Class within any block scope
9	Inheritance	1. Inheritance 2. Extends 3. Supper & Subclass 4. Advantages of Inheritance 5. Extension in existing class 6. Reference a Subclass Object 7. Using super 8. Multilevel Hierarchy 9. Constructors Call 10. Method Overriding
10	Polymorphism	1. Static 2. Dynamic 3. Casting 4. instanceof
MIDTERM EXAM		
12	Abstract Classes & Interfaces	1. Abstract classes 2. Why Abstract class 3. Example 4. Interface 5. Weak Inheritance 6. Difference b/w Abstract class and Interface
13	Graphical User Interface	1. Event Handling 2. Layouts 3. Text Input 4. Choice Components

